

Technical Specification Document

on development of VialApp platform

1. Description of the project

Project concept:

1. VialApp is an online platform designed to help people in Chile prepare for the theoretical driving test. It provides practice tests and study tools for Class B and Class C driving licenses. Users can try some tests for free and, if they want full access to the course, they can purchase a subscription for 7, 15, or 30 days.

Target audience

2. The target audience is people in Chile who are preparing to obtain a Class B or Class C driving license and need to study for the theoretical driving test. VialApp is mainly intended for people who want to practice online and evaluate their knowledge before taking the official test.

How will users learn about the project? What marketing tools are used and through which channels will traffic be attracted?:

3. Users will learn about VialApp mainly through social media and online advertising. Platforms such as Instagram, Facebook, TikTok, and Google Ads can be used to reach people who are preparing for their driving test.

Will they use the project from mobile devices?:

4. Yes. VialApp can be used from mobile devices, allowing users to study and take practice tests directly from their smartphones or tablets. The platform is designed to be accessible through a web browser, so users do not need to install an application.

KPIs of the business:

5. The project launch should mainly influence the number of registered users, the number of paid subscriptions, and the conversion rate from free users to paid users.

technical support of the project required after development:

6. The project requires technical support after development. This includes web hosting, server and database administration, maintenance, bug fixing, security updates, and regular content updates. The platform also needs to be monitored to ensure that payments, subscriptions, and practice tests are working correctly.

Project team:

1. Internal customer, decision maker

The internal customers and decision makers are the owners of VialApp, who decide how the platform should be developed and improved.

Departments involved:

2. The project involves the owners and developers of VialApp. Since it is a small project, there are no separate departments involved.

Content:

3. The content is prepared and managed by the owners of VialApp.

Marketing:

4. Marketing is managed by the owners of VialApp.

Timing:

5. The project should be developed within the planned academic period.

Hypotheses of the project:

We expect that VialApp will be mainly used by people who are preparing for their Class B driving test and want to practice before taking the official exam.

We expect that users who try the free tests and find them useful will be more likely to purchase a subscription.

Success criteria:

The project will be considered successful if the number of registered users and paid subscriptions increases after the launch, and a significant percentage of users who try the free tests purchase a subscription.

2. Sections and Content

Short description:

Similar solutions on the market:

1. There are similar platforms in the Chilean market, such as PracticaTest. VialApp can use these platforms as a reference while focusing on providing a simple and accessible experience for users preparing for their driving test.

Partition structure:

2. The main functions are user registration and login, free practice tests, subscription plans, online payments, access to the full course, and test results.

Connection with other services:

3. VialApp is connected to a database for user accounts, test information, results, and subscriptions. It is also connected to an online payment system.

Content

1. What already exists?

Some initial content, texts, and graphic materials are already available, but they will need to be reviewed and adapted for the project.

Answer:

2. What is missing?

The backend, database, payment system, user accounts, and subscription management still need to be developed.

Answer:

3. Is there a corporate identity or corporate design guidelines?

Yes. VialApp already has a logo and a defined set of colors that will be used for the visual identity of the project.

Structure of the Project

Section List:

Section 1 Home

Section Description:

The main page of VialApp. It will introduce the platform, explain its main features, and allow users to start a free practice test.

Section 2 User Account

Section Description:

Users will be able to register, log in, and manage their account and subscription.

Section 3 Courses

Section Description:

Users will be able to choose between the Class B and Class C driving license courses and access the available study content.

Section 4 Practice Tests

Section Description:

Users will be able to take practice tests and answer questions related to the selected driving license.

Section 5 Results

Section Description:

Users will be able to see their test results and review their correct and incorrect answers.

Section 6 Subscription Plans

Section Description:

Users will be able to view and compare the available 7, 15, and 30-day subscription plans and select the one they want to purchase.

Section 7 Payments

Section Description:

Users will be able to pay for the selected subscription plan online. After a successful payment, the subscription will be activated automatically.

3. Design task

References: PracticaTest.cl – reference for the structure of driving test preparation platforms.

Link to prototype: Not ready yet

Section 1 Home

The mechanics of the block: The user can navigate to the courses, start a free test, register, or log in using the main buttons.

Design requirements: The page should have a simple and modern design, using the VialApp logo and brand colors. The main actions should be easy to find.

Section 2 User Account

The mechanics of the block: The user can register, log in, view their subscription, and access their courses.

Design requirements: The interface should be simple and easy to use on both desktop and mobile devices.

Section 3 Courses

The mechanics of the block: The user can select between the Class B and Class C courses and access the available content.

Design requirements: The two courses should be clearly separated and easy to identify.

Section 4 Practice Tests

The mechanics of the block: The user answers one question at a time and can move to the next question. After completing the test, the results are displayed.

Design requirements: Questions and answers should be easy to read, with large buttons and a layout suitable for mobile devices.

Section 5 Results

The mechanics of the block: After completing a test, the user can see their score and review their correct and incorrect answers.

Design requirements: The results should be presented clearly, making it easy for the user to understand their performance.

Section 6 Subscription Plans

The mechanics of the block: The user can view and compare the available 7, 15, and 30-day subscription plans and select the one they want to purchase.

Design requirements: The plans should be clearly displayed with their duration and price. The purchase button should be easy to find.

Section 7 Payments

The mechanics of the block: After selecting a subscription plan, the user continues to the payment process. After a successful payment, the subscription is automatically activated.

Design requirements: The payment process should be simple and clearly show the selected plan and its price.

4. Development task

Idea of design:

<https://www.figma.com/design/6UE365xbS4ypUdqcAhymqp/Sin-t%C3%ADtulo?node-id=0-1&t=2pH3MOdykmoKGwmz-1>

Layouts:

1. <https://github.com/lastvyandy/VialApp-urfu>

Planned domain:

<https://www.vialapp.com>

Website:

Not available yet.

General requirements

1. Used technology: FastAPI for the backend, React for the frontend, and Swagger for API documentation and testing.
2. Browser support: The site should support the most common desktop and mobile browsers, including Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.
3. Adaptability: enumeration of breakpoints: The site should be responsive and adapt to different screen sizes. The main breakpoints are:

Mobile: up to 767 px

Tablet: 768–1023 px

Desktop: 1024 px and above
4. What languages should the project code be written in?: The client side will be written in JavaScript using React, while the server side will be written in Python using FastAPI.
5. Where will the product be located? (hosting, DNS): The product will be hosted on a web server and will use the vialapp.com domain through a configured DNS service.

6. What payment system is used?: VialApp uses Webpay by Transbank to process online payments with bank cards.

Technical requirements

The project requires web hosting capable of running the FastAPI backend and React frontend. The technology stack includes Python with FastAPI for the backend and JavaScript with React for the frontend.

Configuration Features:

The project requires CORS support so the React frontend can communicate with the FastAPI backend. HTTPS should be used for secure communication, and standard web ports should be used unless the hosting environment requires a different configuration.

Integration

Integration List:

- Online payment gateway
- Email service for account notifications and password recovery
- Database used by the backend
- DNS and hosting services

Access Data:

- API keys, credentials, tokens, and configuration variables will be required to connect to external services.

Integration Features:

- The frontend communicates with the FastAPI backend through REST API requests using HTTPS. External services such as the payment gateway can be integrated through APIs and webhooks.

Section 1 User Account

Section content / description of the included area:

Registration, login, account information, and subscription status.

Description of functionality (with complex design):

Users can create an account, log in, recover their password, and view the status of their subscription.

Additional Information: Authentication data must be handled securely.

Section 2 Practice Tests

Section content / description of the included area: Questions, answer options, navigation between questions, and test progress.

Description of functionality (with complex design): Users can answer questions, move through the test, submit their answers, and receive a final result.

Additional Information: The questions and answers are provided by the backend through the API.

Section 3 Subscription Plans

Section content / description of the included area: Available 7, 15, and 30-day subscription plans

Description of functionality (with complex design): Users can select a subscription plan and continue to the payment process.

Additional Information: The selected plan must be sent to the backend before payment.

Section 4 Payment

Section content / description of the included area: Payment information and confirmation of the selected plan.

Description of functionality (with complex design): The user is redirected to the payment gateway. After payment confirmation, the backend activates the subscription automatically.

Additional Information: The payment result should be verified by the backend before granting access.

Section 5 Results

Section content / description of the included area: Test score, correct answers, incorrect answers, and progress information.

Description of functionality (with complex design): After completing a test, users can review their results and identify the questions they answered incorrectly.

Additional Information: Results can be stored in the database and associated with the user's account.